

## CALS ORI Research Investment Area Seed Grant

*Data Science, Artificial Intelligence and Decision Support Systems — FY27*

### **Causal AI for Real-Time Agricultural Shock Monitoring: Building a Multimodal Counterfactual Forecasting Platform for Policy, Community, and Business Decision Support**

This team brings together five researchers from three colleges, including the Sanghani Center for Artificial Intelligence and Data Analytics, with an established record of collaboration that includes co-authored publications and jointly developed grant proposals.

#### **Team and Budget Information**

| <b>Role</b>               | <b>Name</b>    | <b>Degree</b> | <b>Department / Unit</b>                         | <b>College / Center</b>                                      | <b>Email / Contact</b>          |
|---------------------------|----------------|---------------|--------------------------------------------------|--------------------------------------------------------------|---------------------------------|
| Principal Investigator    | Le Wang        | Ph.D.         | Department of Agricultural and Applied Economics | College of Agricultural and Life Sciences; Kohl Centre       | lewang@vt.edu (primary contact) |
| Co-Principal Investigator | Xuan Wang      | Ph.D.         | Computer Science                                 | Sanghani Center for Artificial Intelligence & Data Analytics | xuanw@vt.edu                    |
| Co-Principal Investigator | Huaiyang Zhong | Ph.D.         | Industrial and Systems Engineering               | College of Engineering                                       | hzhong@vt.edu                   |
| Co-Principal Investigator | Chunbei Wang   | Ph.D.         | Department of Agricultural and Applied Economics | College of Agricultural and Life Sciences                    | chunbeiwang@vt.edu              |
| Co-Principal Investigator | Yang Shao      | Ph.D.         | Geography                                        | College of Natural Resources and Environment                 | yshao@vt.edu                    |

#### ***Team Expertise***

**Le Wang, PI, Agricultural and Applied Economics / Kohl Centre.** Le Wang is an economist and econometrician whose research develops and applies causal inference and machine-learning methods for policy evaluation in agricultural, rural, labor, and public-policy settings. He brings expertise in synthetic control, difference-in-differences, instrumental variables, and causal ML, which are central to estimating the impacts of disasters, policies, and shocks when conventional outcomes are delayed. He will lead the causal identification framework, multimodal synthetic control strategy, policy evaluation design, stakeholder engagement, and future center-building agenda.

**Xuan Wang, Co-PI, Computer Science / Sanghani Center.** Xuan Wang brings expertise in artificial intelligence, data mining, natural language processing, AI for science, limited-supervision learning, large language models, and multimodal scientific foundation models. Her role is central to the project's image-based AI component, including learning representations from remote-sensing imagery, developing counterfactual image prediction methods, and adapting foundation-model approaches to agricultural and disaster-monitoring contexts.

**Huaiyang Zhong, Co-PI, Industrial and Systems Engineering.** Huaiyang Zhong brings expertise in data analytics, disaster prediction, health systems, decision support, and policy-relevant modeling. His prior work on multimodal frameworks for disaster and multi-outcome prediction directly informs the project's goal of real-time causal monitoring under policy and disaster shocks. He will help translate image-based causal estimates into operational decision logic, interactive dashboards, and policy simulators for policymakers, Extension agents, emergency-response coordinators, agribusinesses, and community users.

**Chunbei Wang, Co-PI, Agricultural and Applied Economics.** Chunbei Wang is a labor and public economist whose research spans farm labor, immigration, small-business dynamics, entrepreneurship, public policy, and local economic outcomes. Her expertise is important for connecting remote-sensing and image-based shock

indicators to broader rural economic consequences, including labor-market adjustment, farm-business activity, household responses, entrepreneurship, and community-level outcomes following disasters or policy changes.

**Yang Shao, Co-PI, Geography.** Yang Shao is a geographer with expertise in remote sensing, GIS, image processing, land-cover change detection, spatial statistical modeling, machine-learning-based image classification, crop mapping, yield estimation, and near real-time environmental monitoring. His expertise is essential to the project’s remote-sensing foundation, including acquiring and processing satellite imagery, constructing image-derived agricultural indicators, validating counterfactual image outputs, and ensuring that the AI-generated measures are scientifically credible and policy relevant.

### Prior Collaboration and Team Readiness

The proposed team is collaboration-ready rather than newly assembled for this seed grant. Le Wang has published multiple papers and developed proposals with Xuan Wang, demonstrating an established foundation for cross-disciplinary work at the intersection of economics, AI, and applied policy. He has also published multiple papers with Chunbei Wang, reflecting a sustained collaboration in labor, agricultural, and public-policy research. In addition, Le has worked with Huaiyang Zhong on an internal Policy Fellow grant focused on developing data infrastructure for health-related research, providing direct experience in building applied data systems for decision-relevant analysis. Le has also collaborated with Huaiyang Zhong and Chunbei Wang on a USDA proposal related to community development, and with Chunbei Wang and Yang Shao on a USDA proposal related to yield monitoring, directly aligning with the current project’s emphasis on agricultural monitoring, remote sensing, and policy-relevant data science. These prior collaborations reduce start-up costs, strengthen feasibility within the one-year seed-grant period, and position the team to pursue a large external grant leading toward a future **Center for Multimodal Causal Forecasting and Counterfactual Monitoring**.

### Ongoing Funded and Pending Projects Relevant to Agro-Intelligence Systems

| Role                                      | Funder        | Amount    | Title                                                                                   |
|-------------------------------------------|---------------|-----------|-----------------------------------------------------------------------------------------|
| Le Wang(PI)                               | USDA NIFA     | \$748,412 | Internship in Data Analytics in Agriculture, Community, and Rural Economics (DATA-ACRE) |
| Le Wang(Co-PI)                            | VA Corn Board | \$10,000  | Data-Science for grain marketing decision making                                        |
| Le Wang (PI)<br>Huaiyang Zhong<br>(co-PI) | ISCE          | \$10,000  | Infrastructure, Rural Health, and Economic Development: A U.S. County Study             |

### Budget Request and Justification

**Total request: \$30,000 (\$20,000 from CALS ORI + \$10,000 matching from the Sanghani Center for Artificial Intelligence & Data Analytics).**

**Budget Categories (CALS ORI + Sanghani; one year, 8/1/2026–6/30/2027)**

### Budget Request and Justification

| Budget Source | Amount Requested | Purpose / Justification                                                                                                                                                                                                                              |
|---------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CALS ORI      | \$20,000         | Graduate research assistance and training, seed support for the multimodal data pipeline, cloud computing, remote-sensing data, decision-support translation, and prototype dashboard development, and allowable dissemination costs..               |
| Sanghani      | \$10,000         | Graduate research assistance and training, Foundational AI support for multimodal counterfactual image prediction, foundation-model adaptation, uncertainty estimation experiments, graduate research assistance, and allowable dissemination costs. |
| <b>Total</b>  | <b>\$30,000</b>  |                                                                                                                                                                                                                                                      |

# Proposal Narrative — Causal AI for Real-Time Agricultural Shock Monitoring

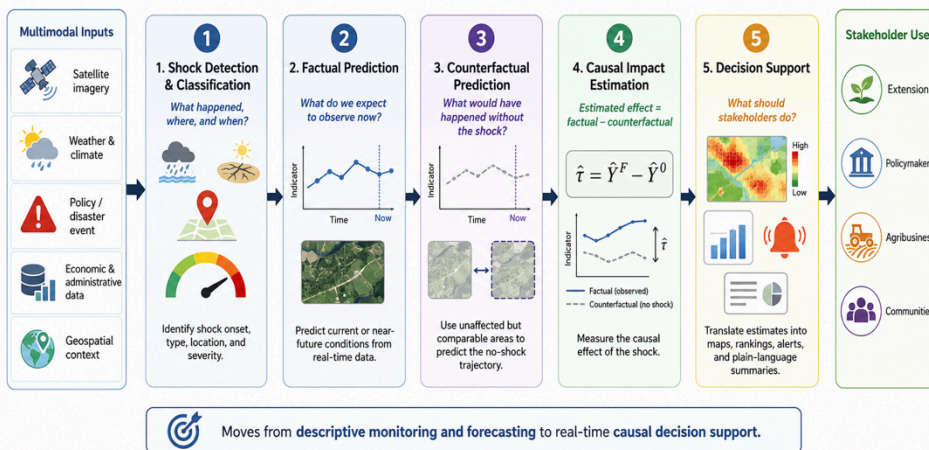
## 1. Project Summary

Agricultural and rural communities increasingly face disruptions from droughts, floods, extreme weather, disasters, market shocks, disease outbreaks, infrastructure failures, and policy changes (IPCC, 2022). Yet the outcomes most important for response and evaluation — crop yields, farm income, employment, business activity, community well-being — are often observed only after substantial delay because traditional data collection processes can take months or even years to compile and release information (Carletto et al., 2015). As a result, decision-makers often must respond before reliable outcome data become available. Agricultural monitoring systems and yield-forecasting tools have advanced substantially in recent years (Jeong et al., 2016; Khaki & Wang, 2019), but they can only tell us what *happened*; modern forecasting can tell us what *may* happen. Neither answers the central causal question decision-makers face for impact evaluation: what *would have* happened had the shock or policy not occurred? (Rubin, 1974; Imbens & Rubin, 2015)? Answering this question requires, first, real-time monitoring techniques to track outcomes as conditions evolve and, second, credible counterfactuals that estimate how outcomes would have evolved had the disruption not occurred (Abadie et al., 2010; Athey & Imbens, 2017).

This seed project develops a **real-time causal forecasting and counterfactual monitoring framework** for agricultural and rural systems. The long-term vision is a multimodal system integrating conventional economic data, administrative records, surveys, remote sensing, mobility data, sensor streams, text, and images (Reichstein et al., 2019; Donaldson & Storeygard, 2016). As the first step, this project focuses on the most technically distinctive and decision-relevant component: **image-based causal inference using remote sensing data**, which is high-frequency, spatially detailed, and often available well before conventional outcomes (Burke & Lobell, 2017; Jean et al., 2016).

The project will build a **five-stage pipeline**: shock detection and classification → factual prediction → counterfactual prediction → causal impact estimation → decision support. The system will first identify the timing, location, type, and severity of a **shock**. It will then monitor and forecast image-based outcomes in affected areas, construct the no-shock counterfactual using comparable unaffected areas, and estimate causal impacts as the difference between observed and counterfactual outcomes (Abadie et al., 2010). Finally, the results will be translated into decision-support tools - including maps, rankings, uncertainty measures, alerts, and plain-language summaries - for stakeholders and policymakers.

### Intuitive Causal Forecasting Pipeline



## 2. Background and Motivation

When a drought, flood, hurricane, disease outbreak, infrastructure disruption, market shock, or policy change affects an agricultural or rural region, decision-makers face a familiar information gap. Official data on yields, farm income, labor demand, employment, business activity, and local economic outcomes often arrive months or quarters after decisions must be made (Carletto et al., 2015; USDA ERS, 2023). Satellite imagery, weather data, sensor streams, mobility indicators, and administrative records can arrive within days, but they

describe current conditions without identifying how much of the observed change was caused by the shock (Burke & Lobell, 2017; Donaldson & Storeygard, 2016).

Agricultural data science has made substantial progress in descriptive monitoring and predictive forecasting — dashboards, remote-sensing indices, crop-condition models, yield forecasts, and weather-driven risk tools (Jeong et al., 2016; Khaki & Wang, 2019). Policy and disaster response, however, require a further layer: **real-time causal decision support**, i.e., credible estimates of how observed conditions differ from the trajectory that would have occurred without the shock (Rubin, 1974; Pearl, 2009; Athey, 2017). This counterfactual is what underlies policy evaluation, disaster assessment, resource allocation, insurance decisions, and business planning.

The PI has developed and applied causal inference and machine-learning methods for conventional structured datasets, including synthetic control, difference-in-differences, instrumental variables, and causal machine learning. This project

extends that agenda to image-based and multimodal outcomes. The methodological challenge is no longer to estimate a counterfactual scalar outcome but to estimate counterfactual image-derived conditions: what affected agricultural or rural areas would have looked like if the shock had not occurred. Once generated, factual and counterfactual predictions can be translated into interpretable indicators — vegetation stress, crop condition, drought severity, flood exposure, land-cover change, yield deviation, and potential local economic disruption (Tucker, 1979; Huete et al., 2002; Burke & Lobell, 2017). Recent advances in remote sensing, multimodal AI, foundation models, and synthetic-control-style imputation make this feasible, yet no integrated framework currently combines them into a causal forecasting pipeline suitable for agricultural decision support (Reichstein et al., 2019; Donaldson & Storeygard, 2016).

### 3. Long-Term Vision and Development Pathway

The long-term vision is a **Center for Multimodal Causal Forecasting and Counterfactual Monitoring** that integrates conventional economic data, administrative records, surveys, remote sensing, mobility data, sensor streams, text, and images into a unified causal-inference and decision-support platform serving policymakers, Extension professionals, emergency-response agencies, agricultural businesses, lenders, insurers, community organizations, and researchers (Reichstein et al., 2019; Donaldson & Storeygard, 2016). The development pathway is deliberately staged: **seed grant** → **large external grant** → **potential center-scale proposal**. The seed establishes proof of concept, team integration, preliminary infrastructure, and a stakeholder-facing prototype. The large external grant expands the framework across shocks, geographies, modalities, and stakeholders. The center institutionalizes the platform as durable infrastructure for agricultural resilience, rural economic development, policy evaluation, and disaster response, mirroring NSF's convergence research model (NSF, 2022). Seed deliverables are designed to support competitive large external grant submissions within 12–24 months. Targets include **USDA NIFA Data Science for Food and Agricultural Systems**, **USDA AFRI Strengthening Agricultural Systems**, **NSF Smart and Connected Communities**, **NSF Collaborations in AI and Geosciences**, **NASA Earth Science Applications**, **NOAA climate and disaster decision-support programs**, and **DARPA** concepts on causal multimodal disaster monitoring. The seed establishes proof of concept and the team-integrated prototype; the large grant will expand the framework across shocks, geographies, modalities, and stakeholders.

### 4. Project Goals and Objectives

**Overall goal:** develop and pilot a causal forecasting and counterfactual monitoring framework that estimates the impact of an agricultural, environmental, policy, or disaster shock from remote-sensing imagery and auxiliary data in near real time, and translates the estimates into stakeholder-ready decision support.

**Objective 1.** Build a reproducible multimodal data pipeline ingesting Sentinel-2, Landsat, MODIS, weather and climate data, land-use layers, geospatial context, and administrative or economic data, supporting county and sub-county analysis (Drusch et al., 2012; Didan et al., 2015). Pilot search will prioritize Virginia and Appalachia but the final event will be selected by data availability, shock severity, image quality, treatment-control variation, and external-funding relevance.

**Objective 2.** Develop a **real-time factual prediction layer** that produces observed-state estimates of vegetation stress, crop condition, flood exposure, drought intensity, land-cover change, surface temperature, soil-moisture proxies, and learned remote-sensing embeddings. We will rely on established vegetation indices such as NDVI and EVI and related MODIS-based measures for vegetation condition and land-surface monitoring (Tucker, 1979; Huete et al., 2002; Didan et al., 2015).

**Objective 3.** Develop a **counterfactual prediction layer using multimodal synthetic control**, applying the logic of synthetic control to image-based and multimodal outcomes — primarily in feature space using interpretable indicators, and exploratorily in latent or image space via foundation-model embeddings (Abadie et al., 2010; Athey et al., 2021; Ben-Michael et al., 2021).

**Objective 4.** Estimate causal impacts as the difference between factual and counterfactual predictions, validating with pre-shock prediction accuracy, placebo tests, leave-one-region-out cross-validation, sensitivity analysis, and comparison against later-released administrative outcomes (Abadie et al., 2010; Chernozhukov et al., 2018; Imbens & Rubin, 2015).

**Objective 5.** Translate estimates into a prototype dashboard, methodological working paper, stakeholder impact brief, and large-grant concept note, supporting submission of a >\$3M interdisciplinary external grant within 12–24 months. This objective aligns with the type of cyberinfrastructure and decision-support tools envisioned in convergence and climate-resilience programs (NSF, 2022; IPCC, 2022).

**5. Proposed Pilot Application** The pilot will focus on a recent drought, flood, or other agricultural or rural shock selected at project start. Selection will be data-driven, evaluating candidate events on shock severity, spatial variation, image availability, cloud cover, pre-shock histories, comparison-area availability, and decision relevance. Virginia and the

Appalachian region are preferred candidate settings given Virginia Tech's land-grant mission and regional stakeholder networks, but the team will not pre-commit to a specific geography before assessing the data; if a stronger empirical case is available elsewhere, we will select the setting that provides the clearest proof of concept. The pilot will instantiate the full five-stage pipeline introduced in the figure of Section 1 (shock detection and classification → factual prediction → counterfactual prediction → causal impact estimation → decision support) on a single, well-chosen event. This makes explicit that the project is not only a disaster-classification model, not only a forecasting exercise, and not only a synthetic-control application, but an integrated causal forecasting system in which prediction supports counterfactual impact estimation and decision-making (Rubin, 1974; Abadie et al., 2010; Athey & Imbens, 2017).

**6. Research Strategy Conceptual framework.** For region  $i$  at time  $t$ , the framework estimates two quantities:  $Y_{it}^F$ , the factual or observed-state prediction using real-time data, and  $Y_{it}^0$ , the predicted no-shock counterfactual. The estimated causal impact is  $\tau_{it} = Y_{it}^F - Y_{it}^0$ . This follows the potential outcomes framework in Rubin (1974) and Imbens and Rubin (2015), with an emphasis on using high-frequency, image-derived measures as outcomes. Here  $Y$  is an interpretable outcome translated from imagery. Prediction is therefore central to the causal framework: factual prediction captures what is observed under current conditions; counterfactual prediction estimates what would have occurred absent the shock; their difference is the causal estimate (Athey, 2017; Chernozhukov et al., 2018).

**Module 1: Real-time factual prediction.** We extract interpretable indicators (NDVI, EVI, NBR, NDMI, land-surface temperature, soil-moisture proxies, fractional cover, flood exposure, crop stress, land-cover measures) and construct learned image embeddings via AI-based representation learning where appropriate (Tucker, 1979; Huete et al., 2002; Didan et al., 2015; Burke & Lobell, 2017). These models provide  $Y_{it}^F$ , the best estimate of observed or near-future conditions given real-time data.

**Module 2: Counterfactual prediction using multimodal synthetic control.** This is the causal core of the project. Synthetic control methods construct counterfactual paths for treated units by optimally weighting donor units to reproduce pre-treatment trajectories and characteristics (Abadie et al., 2010; Athey et al., 2021; Ben-Michael et al., 2021). Rather than relying only on the standard synthetic-control construction of fixed convex weights over donor units, we apply the broader *logic* of synthetic control — that comparable unaffected areas contain information about what would have happened to affected areas absent the shock — and operationalize it through **learned counterfactual-imputation models** that use donor-pool observations, pre-shock trajectories, land use, weather, climate, and image-derived features. As a transparent baseline, we will estimate feature-space synthetic-control models in which unaffected donor regions are weighted to reproduce the affected region's pre-shock trajectories and baseline characteristics. We will then extend this framework with multimodal covariate adjustment to improve prediction of the no-shock trajectory (Chernozhukov et al., 2018).

**Module 3: Image-to-outcome translation and Decision Support.** An image is not itself the stakeholder outcome. A translation layer maps factual and counterfactual image-derived quantities onto decision-relevant indicators (vegetation stress, crop condition, affected acreage, yield deviation, land-cover change) and, via linkage to county-level data, to farm employment, business activity, and local economic conditions (Jean et al., 2016; Burke & Lobell, 2017; Donaldson & Storeygard, 2016). The layer combines established remote-sensing calibrations, supervised models trained on historical indicator-outcome pairs, and economic models linking shock indicators to rural labor and business outcomes. The primary deliverable operates in feature space using interpretable, scientifically validated indicators, making outputs auditable by Extension agents, emergency managers, and agricultural stakeholders (Tucker, 1979; Huete et al., 2002).

**7. Expected Outcomes and Deliverables** A reproducible multimodal data pipeline; real-time factual prediction model; counterfactual prediction model using multimodal synthetic-control-style imputation; image-to-outcome translation layer; pilot case study; prototype decision-support dashboard; methodological working paper; stakeholder impact brief; and a concept note and proposal plan for a large external submission.

## 8. Timeline (8/1/2026 – 6/30/2027)

**Months 1–3:** identify candidate shocks; select pilot; build data pipeline; define treated and comparison areas; construct remote-sensing indicators. **Months 3–7:** develop real-time factual prediction and feature-space counterfactual-imputation models; estimate no-shock counterfactuals; placebo and sensitivity tests; begin validation. **Months 6–10:** exploratory latent/image-space counterfactual prediction with foundation-model adaptation; build image-to-outcome translation layer and dashboard prototype. **Months 9–12:** refine validation; stakeholder brief; working paper; external-grant concept note; CALS ORI presentation; large-grant submission preparation.

## Nonacademic Stakeholder Impact Statement (Appendix)

When a drought, flood, or other shock hits rural Virginia and the broader Appalachian region, the people who must respond — farmers, Extension agents, county emergency managers, agricultural lenders, and small-business owners — face a fundamental information gap. Official outcome data on yields, farm income, employment, and business activity arrive months after the shock, well past the window when response decisions must be made. Satellite imagery and other rapidly available data describe what is happening, but cannot, on their own, tell decision-makers *how much of what they are seeing is attributable to the shock* rather than to ongoing seasonal or economic trends.

This project closes that gap. By estimating, in near real time, what an affected area *would have looked like* had the shock not occurred — and translating the difference into clear, county-level indicators of vegetation stress, crop condition, drought severity, flood exposure, and land-cover change — the framework gives nonacademic stakeholders a defensible quantitative basis for action. Virginia Cooperative Extension agents can target outreach to the counties most adversely affected; emergency managers can prioritize disaster response; agricultural lenders and insurers can make better-informed decisions about credit and claims; community organizations can document need; and farm businesses can plan around verified rather than perceived impact.

The pilot dashboard, co-developed with Extension, is intentionally designed for nonacademic use: maps, plain-language summaries, uncertainty bands, and county-level rollups, rather than raw imagery or technical model outputs. Beyond the pilot, the same framework generalizes to policy changes, market disruptions, and future climate shocks, positioning the proposed Center for Multimodal Causal Forecasting and Counterfactual Monitoring as durable, decision-grade infrastructure for Virginia's agricultural and rural communities.

## References

- Abadie, A., Diamond, A., & Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490), 493–505.<sup>[26]</sup>
- Athey, S. (2017). Beyond prediction: Using big data for policy problems. *Science*, 355(6324), 483–485.
- Athey, S., Bayati, M., Doudchenko, N., Imbens, G. W., & Khosravi, K. (2021). Matrix completion methods for causal panel data models. *Journal of the American Statistical Association*, 116(536), 1716–1730.
- Athey, S., & Imbens, G. W. (2017). The state of applied econometrics: Causality and policy evaluation. *Journal of Economic Perspectives*, 31(2), 3–32.
- Ben-Michael, E., Feller, A., & Rothstein, J. (2021). The augmented synthetic control method. *Journal of the American Statistical Association*, 116(536), 1789–1803.
- Burke, M., & Lobell, D. B. (2017). Satellite-based assessment of yield variation and its determinants in smallholder African systems. *Proceedings of the National Academy of Sciences*, 114(9), 2189–2194. <https://doi.org/10.1073/pnas.1616919114><sup>[28][26]</sup>
- Carletto, C., Jolliffe, D., & Banerjee, R. (2015). From tragedy to renaissance: Improving agricultural data for better policies. *Journal of Development Studies*, 51(2), 133–148.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1), C1–C68.
- Didan, K., Munoz, A. B., Solano, R., & Huete, A. (2015). *MODIS vegetation index user's guide (MOD13 series), Version 3.00, June 2015 (Collection 6)*. Vegetation Index and Phenology Lab, The University of Arizona.<sup>[30][29]</sup>
- Donaldson, D., & Storeygard, A. (2016). The view from above: Applications of satellite data in economics. *Journal of Economic Perspectives*, 30(4), 171–198. <https://doi.org/10.1257/jep.30.4.171><sup>[31][27]</sup>
- Drusch, M., Del Bello, U., Carlier, S., Colin, O., Fernandez, V., Gascon, F., Hoersch, B., Isola, C., Laberinti, P., Martimort, P., Meygret, A., Spoto, F., Sy, O., Marchese, F., & Bargellini, P. (2012). Sentinel-2: ESA's optical high-resolution mission for GMES operational services. *Remote Sensing of Environment*, 120, 25–36.
- Geirhos, R., Jacobsen, J.-H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., & Wichmann, F. A. (2020). Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11), 665–673.
- Huete, A., Didan, K., Miura, T., Rodriguez, E. P., Gao, X., & Ferreira, L. G. (2002). Overview of the radiometric and biophysical performance of the MODIS vegetation indices. *Remote Sensing of Environment*, 83(1–2), 195–213.
- Imbens, G. W., & Rubin, D. B. (2015). *Causal inference for statistics, social, and biomedical sciences: An introduction*. Cambridge University Press.
- Intergovernmental Panel on Climate Change. (2022). *Climate change 2022: Impacts, adaptation and vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press.
- Jean, N., Burke, M., Xie, M., Davis, W. M., Lobell, D. B., & Ermon, S. (2016). Combining satellite imagery and machine learning to predict poverty. *Science*, 353(6301), 790–794.
- Jeong, J. H., Resop, J. P., Mueller, N. D., Fleisher, D. H., Yun, K., Butler, E. E., Shim, K.-M., Gerber, J. S., Reddy, V. R., Kim, S.-H., & others. (2016). Random forests for global and regional crop yield predictions. *PLOS ONE*, 11(6), e0156571.
- Khaki, S., & Wang, L. (2019). Crop yield prediction using deep neural networks. *Frontiers in Plant Science*, 10, 621.

National Science Foundation. (2022). *Growing convergence research (GCR)* (NSF 22-526). National Science Foundation.<sup>[32][33]</sup>

Pearl, J. (2009). *Causality: Models, reasoning, and inference* (2nd ed.). Cambridge University Press.

Reichstein, M., Camps-Valls, G., Stevens, B., Jung, M., Denzler, J., Carvalhais, N., & Prabhat. (2019). Deep learning and process understanding for data-driven Earth system science. *Nature*, 566(7743), 195–204.

Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5), 688–701.

Tucker, C. J. (1979). Red and photographic infrared linear combinations for monitoring vegetation. *Remote Sensing of Environment*, 8(2), 127–150.

United States Department of Agriculture, Economic Research Service. (2023). *Farm income and wealth statistics*. U.S. Department of Agriculture, Economic Research Service.